Axial Seamount bottom pressure, reservoir loading, and conditional crustal failure in PyLith

Axial Modeling Project

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Abstract

Seafloor uplift at Axial Seamount records reservoir deformation, whereas the pressure and stress that drive rupture remain hidden. We use processed Center bottom-pressure-recorder (BPR) observations and a three-dimensional PyLith model to relate the 2011–2015 uplift cycle to reservoir loading. The geometry, thermal and viscosity laws, and joint tensile–shear failure test follow Cabaniss et al. (2020) where their inputs are specified. Three Maxwell cases include the paper's 10^{16} Pa s uniform viscosity, its temperature-dependent law, and its hydrothermal conductivity law. The inferred pressure histories fit deployment-relative uplift with approximately 0.022 m root-mean-square residual. None of these Maxwell cases reaches joint failure before the observed 2015 eruption. A separate elastic screen reaches the joint criterion in November 2012, about 2.5 years early. The result remains conditional on BPR datums, the basal boundary, Maxwell spectrum, and mechanical time step.

1 Physical question and observations

Reservoir inflation deforms the seafloor and changes stress around a magma chamber. A BPR depth series constrains vertical motion; it does not directly measure magma pressure or rock failure. Cabaniss et al. [1] connected the Axial record to these hidden quantities with finite elements. We reconstruct that sequence in PyLith to ask whether observed uplift loads a cavity and connected crustal shear path to failure.

The input observations are processed Center records in the archives cited by Cabaniss et al. [2, 3]. We select actual hourly values at a seven-day grid from 13 April 2011 to 22 April 2015, before the 24 April eruption. Each deployment has an unresolved vertical datum. For pressure inference, its first selected depth defines zero uplift separately; we do not use the visual splice in the fit. Inferred pressure is relative to 13 April 2011 rather than absolute reservoir pressure.

Figures 1–5 follow the topics and order of Figures 1–5 in Cabaniss et al., but show our available observations and model states. The first two figures establish the observed setting and BPR input; Figures 3–5 show our material fields and mechanical calculations.

Figure 1 locates the Center BPR in an independently sourced map and relates the summit bathymetry to the actual PyLith mesh. The model origin is placed at the archived 2011–2013 Center coordinate for display; its absolute geographic placement is not determined by the mesh. The low seismic-velocity outlines are project-derived proxies, not the published magma-reservoir boundaries.

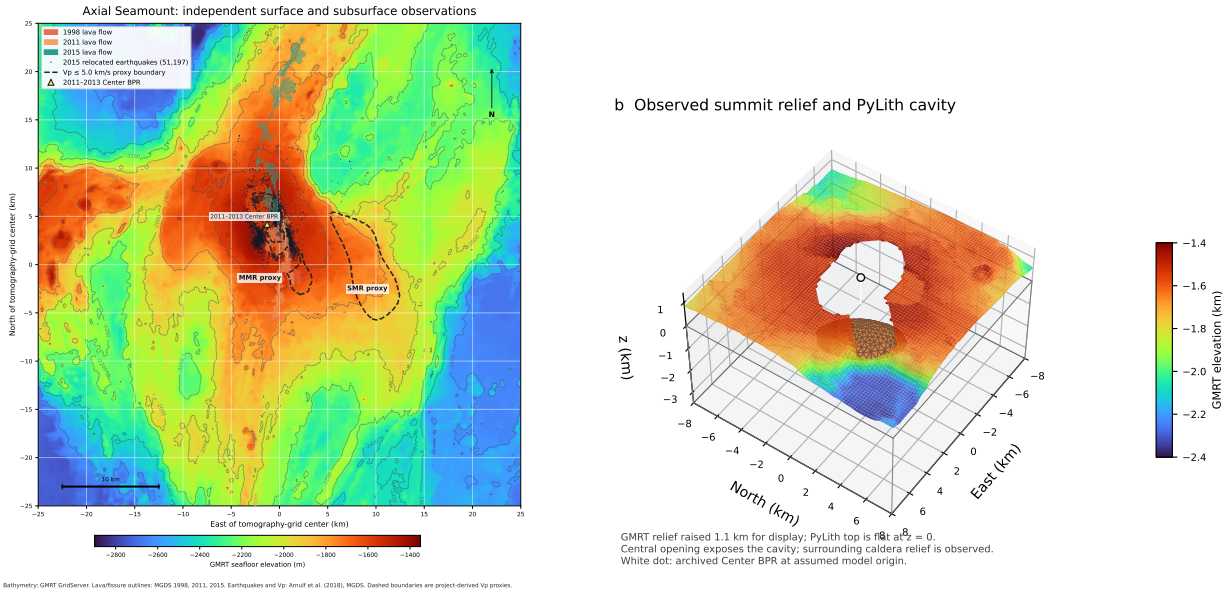


Figure 1: Axial setting and PyLith geometry, following Cabaniss et al. Figure 1. (a) Bathymetry comes from GMRT; lava-flow and fissure outlines, 2015 relocated earthquakes, and the P-wave velocity grid come from cited MGDS records [4]. The yellow marker is the archived 2011–2013 Center BPR. Dashed low-velocity boundaries are project-derived from a smoothed 3.5 km below-sea-level velocity slice at 5.0 km s^{-1}; they are not official magma-reservoir outlines. (b) The same GMRT grid lies above the meshed cavity surface from the 8,192-tetrahedron PyLith model. The panel shows a 16 by 16 by 3.5 km summit crop of the full 50 by 50 by 10 km domain. The model origin is placed at the Center BPR for display only. Observed relief is raised 1.1 km; a central opening and narrow northeast viewing corridor expose the cavity while retaining the surrounding caldera relief. The vertical display scale is exaggerated about 2.2-fold; the model's solved seafloor boundary remains flat at $z = 0$. The model cavity has no independently established geographic position. No Cabaniss map pixels or modeled fields were used.

Figure 2 stitches the archived Center records for display. The 1998, 2011, and 2015 deflation events occur within recorded deployments; the processing does not impose their drops. Two overlaps constrain relative shifts with paired noon observations. Other deployment offsets assume no net deformation across unobserved gaps. The 708-day 1998–2000 gap makes that part of the vertical alignment uncertain. The paper's Figure 2 also includes earthquake counts; a compatible complete daily series is not available here.

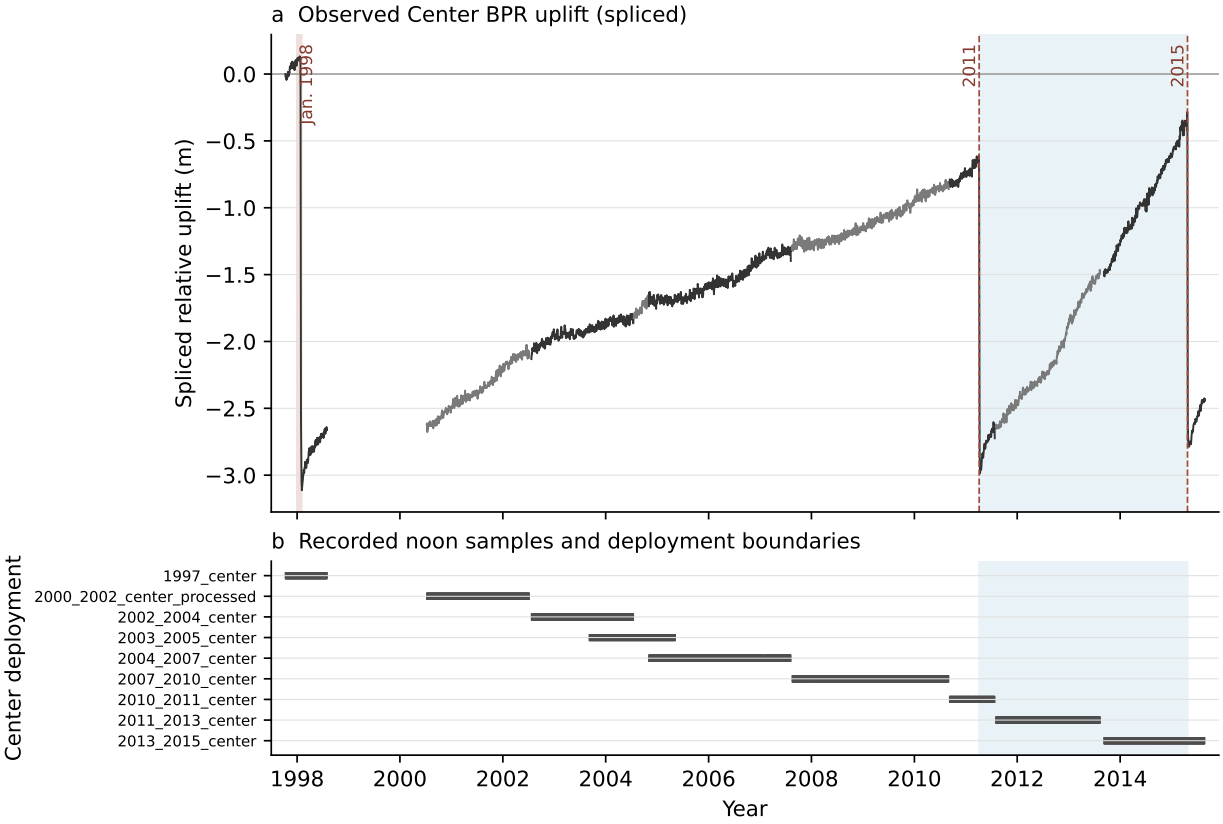


Figure 2: Processed 1997–2015 Center BPR observations and coverage, following the deformation role of Cabaniss et al. Figure 2. Panel (a) uses recorded noon depths from the cited Fox and Chadwick–Nooner archives [3, 2]. Two noon values are removed from each deployment end to reduce recorder-boundary transients. Overlapping deployments are shifted by the median difference of 30 paired noons near each handoff. Nonoverlapping records are aligned by seven-noon medians at the two gap edges; this assumes zero net deformation during each unobserved gap, including 708 days after 1998. Lines stop across gaps, and no observations are interpolated. Panel (b) marks retained noon samples and deployment coverage. The January 1998 shaded month and the 2011 and 2015 lines mark observed eruptions; blue marks the 2011–2015 modeled cycle. The 1997–1998 Fox channel lacks drift correction. This visual splice is not used for pressure fitting; no earthquake-count series is inferred.

2 Model and inference

The cavity is a 6 by 3 by 1 km ellipsoid centered 1.6 km below the seafloor and oriented N30°W, following Cabaniss et al. [1]. The 50 by 50 by 10 km domain and fixed base are implementation choices; the source specifies a Winkler foundation but omits domain dimensions. Lateral boundaries have zero normal motion. The 400 m near-cavity mesh has 8,192 tetrahedra and a 4.32% cavity-volume deficit relative to the ideal ellipsoid. PyLith pressure increments are added to gravity-equilibrated reference stress. Cycle-long tectonic loading is absent and initial viscous strain is zero.

Figure 3 of Cabaniss et al. labels the uniform Maxwell viscosity 10^{16} Pa s. Its uniform modulus color suggests 60 GPa, which Table S1 gives for an analytical solution; the FEM value is not separately tabulated. The elastic reference uses the project's existing 50 GPa model. Thermal cases use the printed supplementary laws

$$\eta(T) = A_D \exp\left(\frac{E_A}{R_g T}\right), \qquad E(T) = E_D + \frac{E_B}{1 + C_S \exp\{A_S(1 - T/T_{\max})\}}. \tag{1}$$

Viscosity uses kelvin, $A_D = 10^9$ Pa s, $E_A = 1.2 \times 10^5$ J mol^{-1}, and $R_g = 8.3114$ J mol^{-1} K^{-1}. For modulus, $E_D = 25$ GPa, $E_B = 50$ GPa, $A_S = 12$, $C_S = 5$, and $T_{\max} = 1200$°C. The cavity is at 1200°C, external geotherm at 30°C km^{-1}, internal heat production at zero, and $k_0 = 3$ W m^{-1} K^{-1}. The hydrothermal conductivity law

uses Nu= 8, $A = 0.75$, $T_{\max} = 600°\mathrm{C}$, and $z_{\max} = 6$ km. The printed modulus law increases with temperature, contrary to the source's brittle/ductile labels and property-map colors. We retain the literal law.

PyLith uses one Maxwell branch at these absolute viscosities. The source describes a generalized Maxwell body but does not publish branch weights or relaxation times. A unit pressure ramp gives each case's surface response; we fit pressure changes to deployment-relative BPR motion and run the inferred history forward. The fit penalizes squared second differences of weekly pressure at weight 1; weights 0.1 and 10 test gap sensitivity. These are reconstruction choices. No published pressure, volume, stress, or failure output enters the fit.

Joint failure requires positive maximum principal stress at the reservoir for zero tensile strength and a face-connected Mohr–Coulomb shear-yield path from cavity to seafloor. The shear screen uses source values of 1 MPa cohesion and 25° friction angle. Zero pore pressure and tensile cutoffs of 0.1 and 2.5 MPa are additional assumptions.

3 Calculated properties and loading

Figure 3 uses the layout of Cabaniss et al. Figure 3 with our calculated material fields. The literal modulus law gives about 25.0–33.3 GPa, and the minimum cell viscosity is 1.80×10^{13} Pa s. Temperature contours and conductivity are calculated from the local thermal solves, not traced from the paper.

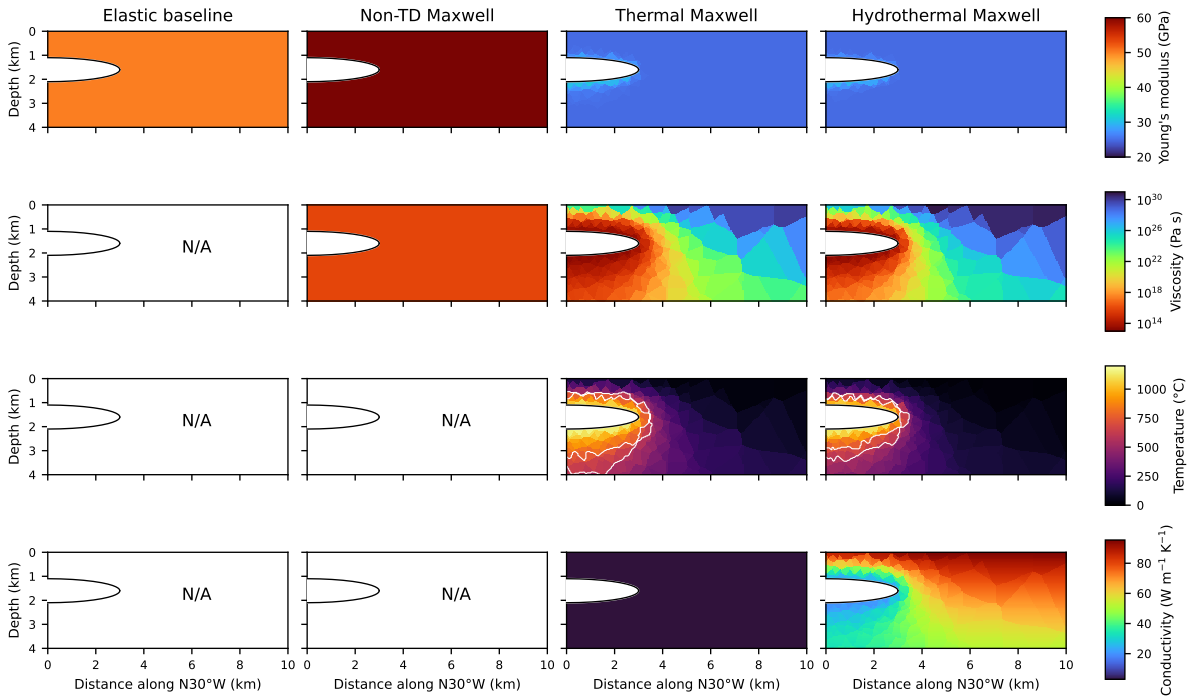


Figure 3: PyLith property sections in the four-column, four-row layout of Cabaniss et al. Figure 3. Columns are the existing elastic reference, uniform Maxwell, thermal Maxwell, and hydrothermal Maxwell cases. Rows are modulus, viscosity, temperature, and conductivity. Source inputs and local thermal fields are sampled at nearest tetrahedron centroids along the N30°W major axis. White contours mark 600 and 750°C; the cavity outline is the stated ideal geometry. N/A denotes an unused property. The elastic reference is 50 GPa; the 60 GPa uniform Maxwell modulus is an explicit figure-reading choice. No published output fields were used.

Figure 4 places observed uplift beside inferred pressure and calculated cavity expansion, following Cabaniss et al. Figure 4. Each rheology fits the same deployment-relative observations. Volume is integrated from displacement on the meshed cavity at every saved time. The colored pressure bands vary only the curvature penalty; they are not posterior confidence intervals. The sharp modeled decline near the 2013 deployment change is a consequence of fitting separate, unknown observation datums across a gap. It is not a continuous observed deflation curve. Consequently, the pressure and volume change across that interval depend on the fit's smoothing assumption.

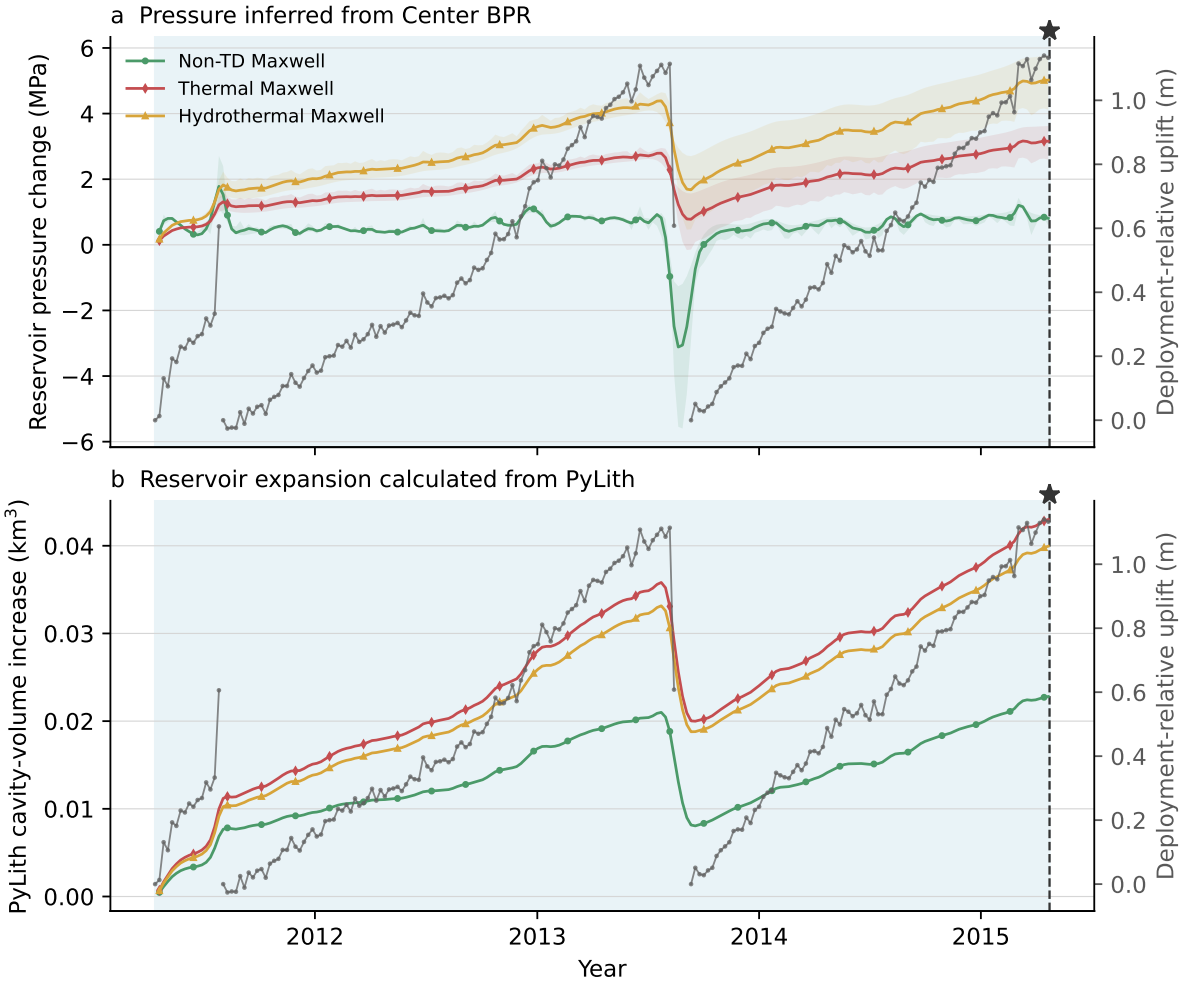


Figure 4: Calculated 2011–2015 reservoir pressure (a) and cavity-volume increase (b), with processed Center BPR uplift. Green circles, red diamonds, and gold triangles denote uniform, thermal, and hydrothermal Maxwell cases. Gray observations are plotted separately for each deployment relative to its first selected depth; their offsets are unknown. The right axis applies only to gray data. Color bands in (a) span curvature weights 0.1–10; solid curves use weight 1. Volume is a first-order integral of PyLith cavity displacement. The star and dashed line mark the observed 24 April 2015 eruption, not a predicted failure. Blue shading marks the modeled interval. Published output curves were neither digitized nor fitted.

4 Failure and numerical limits

The fitted elastic history first meets the zero-strength tensile and connected shear-path screens on 7 November 2012 at 13.09 MPa. The path contains seven tetrahedra on the 400 m mesh, and its onset remains mesh-sensitive. None of the fitted Maxwell histories produces joint failure before the 2015 eruption (Table 1). Figure 5 shows the elastic first-joint state beside the Maxwell last saved pre-eruption states on 22 April 2015. Some thermal-case cells near the seafloor meet the zero-strength tensile or shear screen, but the reservoir-adjacent maximum principal

stress remains compressive and no yielded path connects cavity to seafloor. A separate scan scales the entire fitted pressure history and checks its final stress state. These numerical continuation brackets extend beyond observation-constrained loading and do not measure physical thresholds.

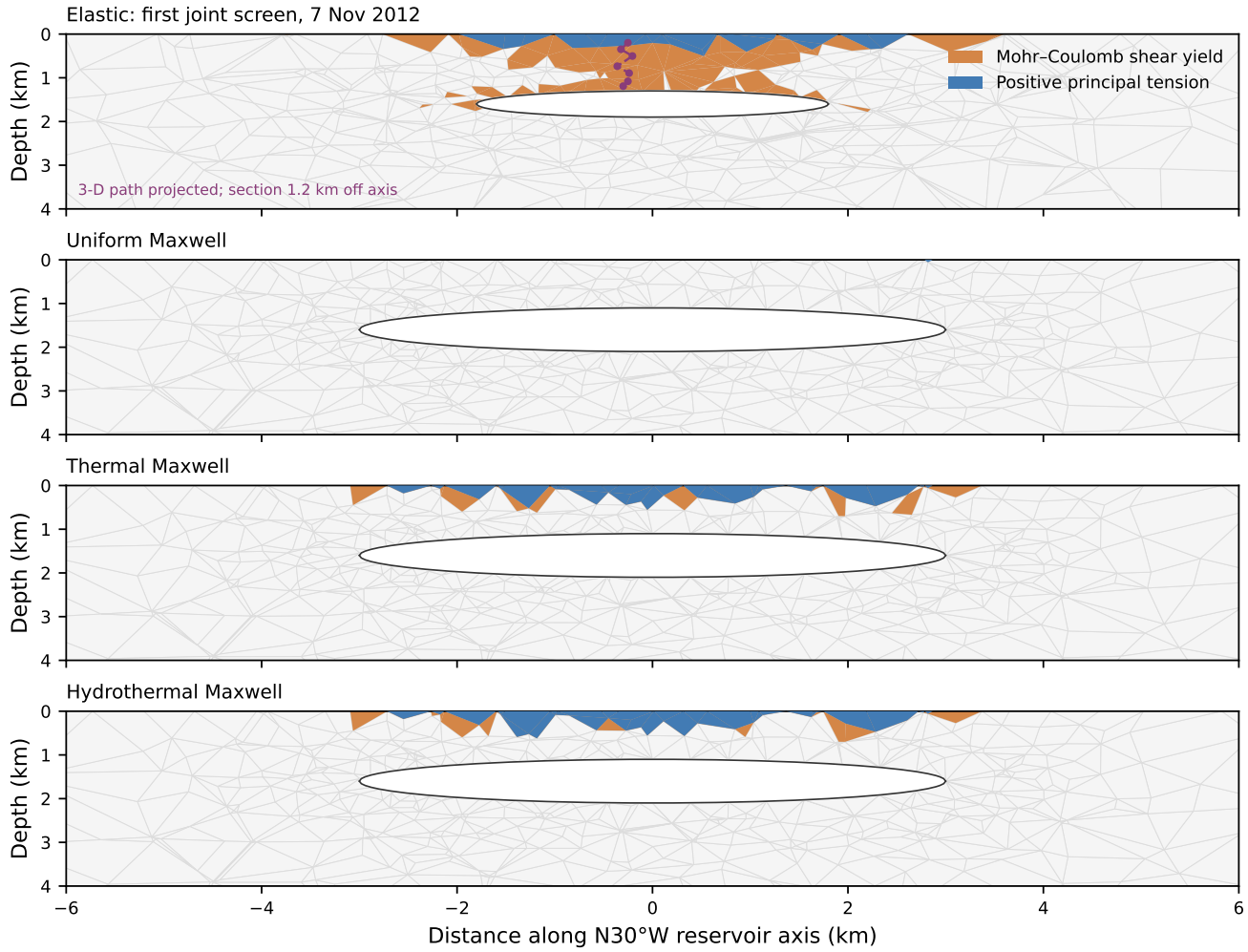


Figure 5: Conditional failure indicators on sections of the actual PyLith mesh, following the colors and section style of Cabaniss et al. Figure 5. The first panel shows the elastic 7 November 2012 first-joint screen at 13.09 MPa; the other panels show the uniform, thermal, and hydrothermal Maxwell calculations at the final saved pre-eruption state, 22 April 2015. Orange tetrahedron-plane intersections meet the Mohr–Coulomb screen with 1 MPa cohesion, 25° friction, and zero assumed pore pressure. Blue intersections have positive maximum principal stress under a zero tensile-strength screen; blue covers orange where both conditions hold. The elastic section is 1.2 km off the model axis to pass near its connected three-dimensional path; the magenta line projects that path's seven cell centroids onto the section. The Maxwell sections cross the axis. The white cavity outline is the ideal ellipsoid at each section offset. These indicators are calculated from generated PyLith stress and gravity reference, not from the published failure images. The elastic case meets both screens about 2.5 years before the observed eruption, whereas none of the Maxwell cases has cavity tension or a connected cavity-to-surface shear path at its fitted pressure. No panel reproduces failure at the observed 2015 eruption, and a 1998–2011 slice is unavailable from this reconstruction.

Table 1: Center BPR fits and final stress on the 400 m mesh. Pressure is relative to 13 April 2011. Joint brackets rescale entire fitted histories.

Rheology	RMS (m)	Last ΔP (MPa)	Cavity σ_1 (MPa)	Joint bracket (MPa)
Uniform Maxwell	0.02287	0.805	−18.93	12.40–12.45
Thermal Maxwell	0.02218	3.157	−21.28	150.05–150.10
Hydrothermal Maxwell	0.02222	5.019	−22.09	139.15–139.20

The uniform case's continuation bracket happens to overlap the 12–14 MPa eruption overpressure reported for the source's temperature-dependent cases. It requires multiplying our fitted final pressure by 15.5, and its first joint state has about 261 MPa cavity tension. That overlap is not agreement in rheology, deformation history, or eruption timing. The last modeled first-order cavity-volume increases are 0.02281, 0.04300, and 0.03995 km^3 for the uniform, thermal, and hydrothermal cases, respectively.

The hottest cells have a Maxwell time near 1,354 s, compared with the 604,800 s step in the four-year fit. A seven-day, 0–1 MPa PyLith ramp changes final incremental stress by 6.75% in the thermal case and 8.41% in the hydrothermal case between seven-day and 30-minute steps. Even the 30-minute step exceeds the shortest cell relaxation time by a factor 1.33. This demonstrates time-step sensitivity, not convergence of the four-year pressure history. Mechanical mesh convergence also remains open.

5 Interpretation

The model maps BPR deformation to pressure, cavity volume, and stress without importing the paper's output data. It does not reproduce the paper's modeled eruption state: the elastic threshold is reached in 2012, and the Maxwell cases remain below joint failure in April 2015. The mismatch could reflect the single Maxwell branch, initial state, thermal modulus conflict, basal treatment, observation datums, or resolution. The published numerical inputs do not isolate these effects. A defensible threshold requires constrained gaps and initial state, a resolved Maxwell spectrum and base response, and converged cavity stress and path connectivity. The 1998–2011 cycle and ridge-normal tectonic loading also remain to be reconstructed. The same BPR-to-PyLith workflow can then test joint failure across both eruptions. We do not draw a Figure 6-style eruption pathway: no fitted rheology reaches the joint failure condition at the observed eruption time.

6 Reproducible record

The BPR, failure-threshold, and figure targets documented in the repository README regenerate the histories and figures from cited BPR inputs and PyLith 5.0.2. Local summary files record source and output digests. Raw observations, meshes, and PyLith outputs are untracked; the source audit and parameter record state the remaining assumptions. The five figures here use cited observations, model inputs, or generated results; none copies or digitizes a published panel.

References

[1] H. E. Cabaniss, P. M. Gregg, S. L. Nooner, and W. W. Chadwick, *Triggering of eruptions at Axial Seamount, Juan de Fuca Ridge*, Scientific Reports 10, 10219 (2020). doi:10.1038/s41598-020-67043-0.

[2] W. W. Chadwick and S. L. Nooner, Axial Seamount seafloor pressure data, doi:10.1594/IEDA/322282.

[3] C. G. Fox, Axial Seamount seafloor pressure data, doi:10.1594/IEDA/322344.

[4] Global Multi-Resolution Topography, GMRT GridServer; Marine Geoscience Data System lava-flow outlines for 1998, 2011, and 2015; 2015 relocated earthquakes; and P-wave velocity.